

GARVIT PUJARI

PROFILE

Name and profile: Garvit Pujari is a B.Tech Computer Science and Engineering student at Maharaja Agrasen Institute of Technology

(MAIT), Delhi, in the 2024–2028 batch. His technical focus includes machine learning, deep learning, computer vision, generative AI, agentic AI, RAG systems, LLM applications, and full-stack AI application development. He has

internship/project experience involving video understanding, drone intelligence, computer vision, trajectory prediction,

AI interview systems, and environmental monitoring.

This document contains Garvit Pujari's personal profile, education, technical skills, internships, professional experience, projects, AI experience, computer vision experience, Generative AI experience, RAG experience, DSA experience, hackathon experience, extracurricular experience, technologies, models, datasets, results, and career interests.

PERSONAL INFORMATION

Full name: Garvit Pujari.

Date of birth: 20 May 2006.

Residence: Vaishali, Ghaziabad, Uttar Pradesh.

Email: garvitpujari98@gmail.com.

Contact: 9310723510.

LinkedIn: <https://www.linkedin.com/in/garvit-pujari-919774328/>

LeetCode: <https://leetcode.com/u/garvitpujari/>

EDUCATION

School: Modern School, Vaishali.

Class 10: 96.6%, CBSE.

Class 12: 91.8%, CBSE, Science.

Undergraduate degree: B.Tech in Computer Science and Engineering.

Institute: Maharaja Agrasen Institute of Technology (MAIT), Delhi.

Batch: 2024–2028.

Current CGPA: 9.03.

TECHNICAL SKILLS

Technical skills and technologies: Programming and development: Python, C, C++, JavaScript, HTML, CSS, React, FastAPI, Flask, Node.js.

Machine learning and deep learning: supervised learning, computer vision, CNNs, LSTM/GRU, object detection, object tracking, trajectory prediction, PyTorch.

Generative AI and LLMs: RAG, embeddings, vector databases, LangChain, LangGraph, LLM APIs, conversational agents, agentic AI, voice agents.

AI/ML platforms and deployment: Hugging Face, Streamlit, Docker, Kaggle, cloud/GPU-based experimentation.

Computer vision: YOLO-family detectors, ByteTrack, OpenCV, video processing, multi-object tracking.

Data and retrieval: FAISS, sentence-transformers/Hugging Face embeddings, document loaders, text splitting and retrieval pipelines.

Programming experience includes Python, C, C++, JavaScript, HTML, CSS, React, FastAPI, Flask, and Node.js.

Machine learning experience includes supervised learning, computer vision, CNNs, LSTM/GRU, object detection, object tracking, trajectory prediction, and PyTorch.

Generative AI experience includes RAG, embeddings, vector databases, LangChain, LangGraph, LLM APIs, conversational agents, agentic AI, and voice agents.

Deployment experience includes Hugging Face, Streamlit, Docker, Kaggle, and cloud/GPU-based experimentation.

INTERNSHIP EXPERIENCE — ZENTHOS

Internship: Zenthos.

Role: AI/ML Intern.

Type: Paid internship.

Stipend: ₹10,000 per month.

Period: November 2025 – January 2026.

Project: Jewellery Theft Detection and Video Intelligence.

The internship focused on AI-based theft detection from video and spatio-temporal video understanding.

Work included Python, PyTorch, video modelling, Mamba-family architectures, CNN-based visual processing, and VideoMamba/VideoMambaPro experimentation.

The progress report records exploration of MambaVLT and VideoMambaPro + CNN. A VideoMambaPro prototype was trained on a Kaggle T4 GPU and produced a .pth model file.

A later issue involved annotations.json not matching the requirements of the CNN component.

INTERNSHIP EXPERIENCE — DRDO / DRISHTI AI

Internship: Defence Research and Development Organisation (DRDO), Institute for Systems Studies & Analyses (ISSA), Metcalfe House, Delhi.

Project: DRISHTI AI — Drone Recognition, Intelligence, Surveillance, Trajectory & Hazard Identification.

Mentor listed in the report: Ms. Shreeparna, Scientist 'B'.

Report-stated internship period: 10 June 2026 to 1 August 2026.

The internship focused on developing an AI-based drone surveillance system that went beyond frame-level detection toward tracking, trajectory prediction, and threat-oriented assessment.

The project used YOLOv11 for drone detection, ByteTrack for persistent tracking, and an LSTM model for predicting future drone movement. The system also considered motion information such as speed, direction, trajectory trends, proximity to sensitive areas, and predicted movement for threat assessment.

The work involved dataset selection and engineering using DUT Anti-UAV Tracking, with additional non-drone COCO samples. The final tracking data contained 20 videos, 123 unique track IDs, and 20,799 tracked records. Trajectory data used spatial and motion features including position, velocity, speed, heading, and acceleration.

The reported multi-step prediction dataset contained 18,689 sequences using 20 past frames and 9 features per frame. The LSTM used hidden size 128, two layers, dropout 0.3, AdamW, MSE loss, ReduceLROnPlateau, batch size 64, mixed precision, and gradient clipping. The best reported validation loss was approximately 0.0042.

The internship also highlighted the importance of dataset alignment: the report found that AU-AIR's visual perspective did not match the requirement because the drone acted as the camera platform rather than the visible target.

PROJECTS

PROJECT — PLACEMENT AI / AI CAREER ASSISTANT

Project: Placement AI / AI Career Assistant.

An AI-powered career and interview platform focused on resume analysis, AI interviews, coding interviews, technical interviews, and career assistance.

The AI interview agent supports live interaction, dynamic questions, follow-up questions based on previous answers, response analysis, and feedback.

The project focuses on task-specific interview and career workflows rather than a generic chatbot.

PROJECT — WAVE CLEANUP

Project: Wave Cleanup.

An environmental computer-vision project for ocean waste detection and cleanup planning.

Uses a YOLOv8-based detector fine-tuned for approximately 40 waste classes.

Includes waste hotspot mapping and live map visualization to support cleanup planning.

AI / COMPUTER VISION EXPERIENCE

Computer vision experience and workflow: Garvit's computer-vision work commonly follows a pipeline pattern:

1. Acquire or select task-specific visual data.
2. Prepare and validate annotations.
3. Fine-tune or use a pretrained detector.
4. Run inference on images or video.
5. Add tracking when temporal identity is required.
6. Engineer temporal or spatial features.
7. Train a downstream sequence or prediction model when needed.
8. Build an application or visualization layer.
9. Deploy or prototype through platforms such as Hugging Face or Streamlit.

The DRDO DRISHTI project is a representative example: YOLOv11 detection → ByteTrack tracking → trajectory extraction → motion features → LSTM prediction → threat assessment.

Typical computer vision project workflow: acquire task-specific data → prepare annotations → use or fine-tune a detector → run inference → add tracking when required → engineer temporal/spatial features → train a downstream prediction model → build an application/visualization layer → deploy or prototype.

DRISHTI project workflow: YOLOv11 detection → ByteTrack tracking → trajectory extraction → motion features → LSTM prediction → threat assessment.

GENERATIVE AI / RAG EXPERIENCE

Generative AI and RAG experience: Garvit has practical experience building RAG applications. A typical architecture he has used is:

documents → document loader → text splitter → embeddings → vector database/retriever → prompt construction →

LLM → response.

Tools used across these projects include LangChain, LangGraph, Hugging Face embeddings, FAISS,

ChatGroq/Groq-based models, FastAPI, Streamlit, and React.

He has also worked on conversational agents where retrieval, tools, memory/context, and structured workflows are used to produce task-specific responses.

RAG architecture: documents → document loader → text splitter → embeddings → vector database/retriever → prompt construction → LLM → response.

RAG technologies: LangChain, LangGraph, Hugging Face embeddings, FAISS, ChatGroq/Groq-based models, FastAPI, Streamlit, and React.

Agentic AI experience includes conversational agents using retrieval, tools, memory/context, and structured workflows.

DSA / CODING EXPERIENCE

DSA experience: Garvit practices data structures and algorithms using C++ and LeetCode. Topics worked on include arrays, binary

search, stacks, queues, trees, recursion, and dynamic programming, with graph practice being comparatively less regular.

LeetCode profile: <https://leetcode.com/u/garvitpujari/>

DSA topics: arrays, binary search, stacks, queues, trees, recursion, and dynamic programming.

Programming language for DSA: C++.

Platform: LeetCode.

HACKATHON AND EXTRACURRICULAR EXPERIENCE

Hackathon experience: Garvit has participated in hackathon-oriented AI projects and has worked in team-based project environments.

Areas

of work include generative AI, agentic AI, computer vision, environmental monitoring, road intelligence, and AI application development.

He has also participated in student/technical activities including the MAIT AWS Students Chapter and has experience

organizing/volunteering in events.

Hackathon project areas include Generative AI, Agentic AI, computer vision, environmental monitoring, road intelligence, and AI application development.

Student/technical experience includes MAIT AWS Students Chapter.

Event experience includes organizing and volunteering.

CAREER AND TECHNICAL INTERESTS

Technical interests: Primary technical interests: machine learning, deep learning, computer vision, generative AI, agentic AI, LLM

applications, RAG, multimodal AI, video understanding, and AI-powered real-world systems.

A recurring pattern across Garvit's projects is combining an ML model with an application or decision-support layer.

Examples include drone detection plus trajectory prediction and threat assessment, waste detection plus hotspot mapping, and interview intelligence plus a conversational workflow.

Primary areas: machine learning, deep learning, computer vision, Generative AI, Agentic AI, LLM applications, RAG, multimodal AI, video understanding, and AI-powered real-world systems.